

AEROTHON 2026 - HAL x IITI

Problem Statement Title : Physics-Informed Digital Twin for Real-Time Four-Stage Turbojet Health Monitoring

Team Name: **VayuCops**

Participants:

Member 1: Divyanshi (Team Leader)

Member 2: Vishal Raj

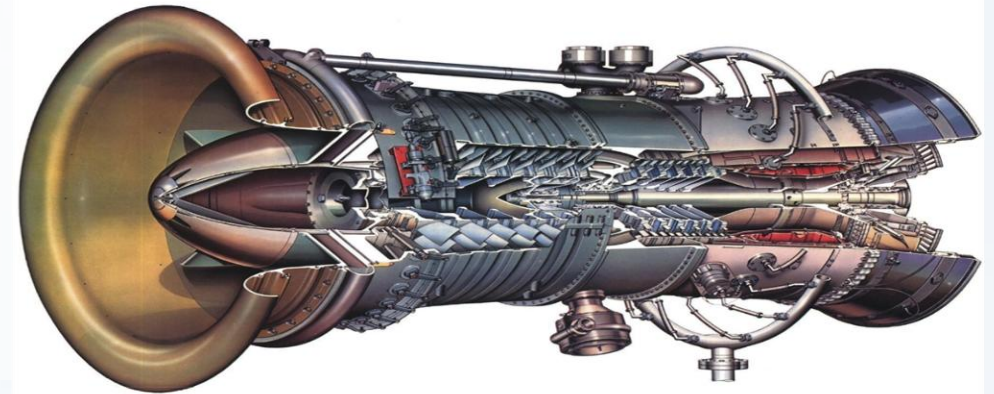
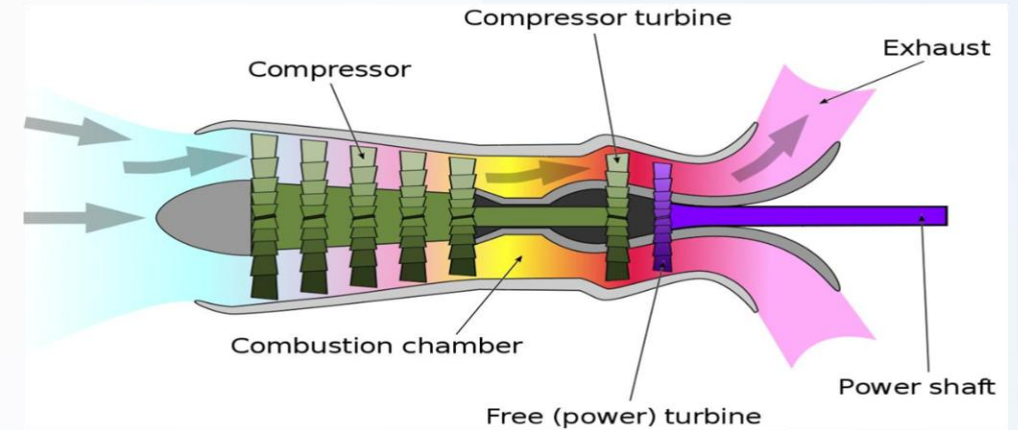
Member 3: Vaishnavi Jaiswal



PROBLEM UNDERSTANDING & MOTIVATION

💡 Why Engine Health Can't Wait for the Next Inspection

- 🕒 Compressor fouling, turbine erosion, and combustor efficiency loss develop gradually — but the parameters that would reveal them can't be measured directly in-flight.
- 🕒 Scheduled inspections catch degradation only after it has already cost fuel efficiency and thrust margin.
- 💻 Full physics-based engine simulations are accurate, but computationally too expensive for continuous, onboard use.



“ *Estimate hidden engine health states from limited sensor data — fast enough to run continuously, accurate enough to trust.* ”

LITERATURE REVIEW / EXISTING APPROACHES

💡 Three Ways People Have Tried to Solve This — and Where Each Runs Out of Road



Model-Based Gas Path Analysis

Kalman filter / linear estimators

Uses thermodynamic engine models with Kalman filtering to estimate component health from gas-path measurements.

Physically grounded and interpretable

Linearized around one operating point — degrades under transient conditions



Pure Data-Driven Models

LSTM / CNN on sensor sequences

Common on benchmark datasets such as NASA C-MAPSS for remaining-useful-life prediction.

Captures nonlinear degradation patterns directly from data

No physical constraints; can violate thermodynamics and generalizes poorly



Physics-Informed Neural Networks

PINNs

Embed governing equations — energy/mass balance, Brayton-cycle relations — directly into the training loss.

Stays physically consistent even with limited data

Mostly shown on single-component or simplified-cycle problems



PROPOSED METHODOLOGY & TECHNICAL APPROACH

💡 Why We Didn't Force One Model to Do Everything

➤ Physics-informed feature engineering

Ten gas-path-derived features — pressure ratios, temperature ratios, temperature/pressure drops, fuel-per-RPM — encode compressor, combustor, and turbine physics directly into the input space, rather than asking the model to discover these relationships from raw pressures and temperatures alone.

➤ Multi-model benchmarking

Eight regression algorithms — Linear and Ridge Regression, Random Forest, Extra Trees, Gradient Boosting, XGBoost, LightGBM, and CatBoost — were evaluated on every target using MAE, RMSE, R^2 , and training time.

➤ Best-model-per-target selection

Rather than forcing one algorithm across all six outputs, each target keeps whichever model performed best for it — Extra Trees for Compressor, Combustor, and Overall, Health; Ridge Regression for Turbine Health; CatBoost for Thrust and TSFC.

OUTPUT OF THIS STAGE

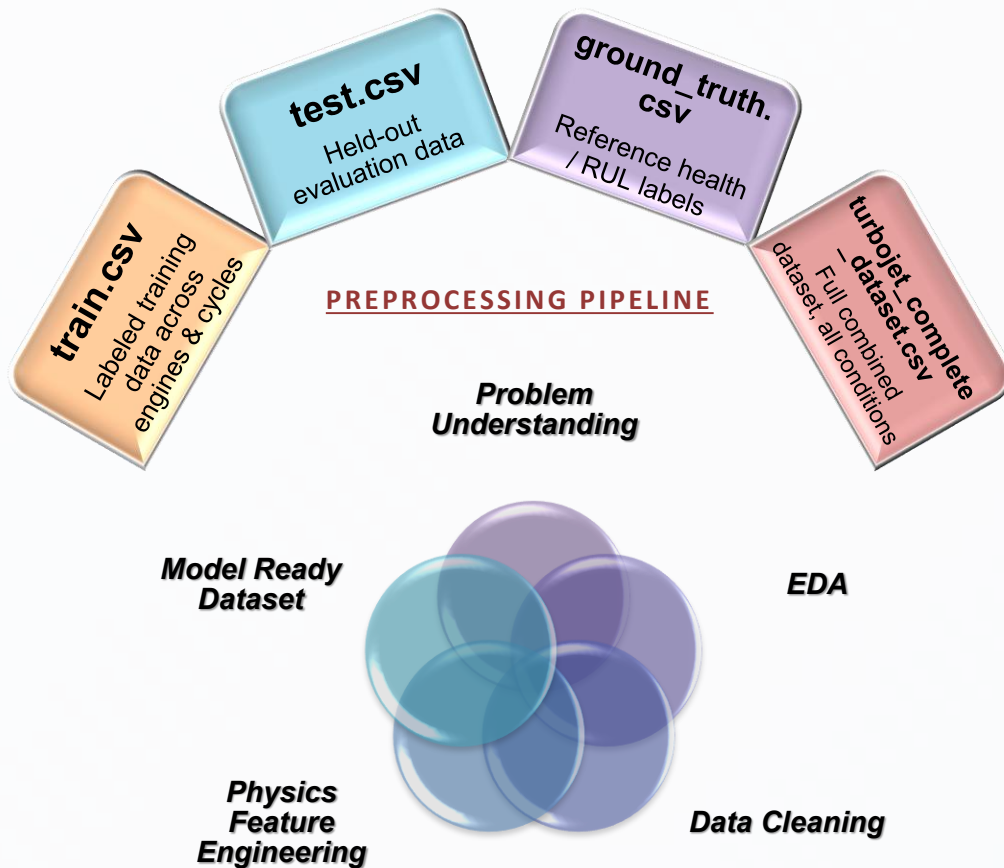
Six calibrated predictions — Compressor, Combustor, Turbine, and Overall Health, plus Thrust and TSFC — feeding into AeroTwin's explainability, uncertainty, and maintenance layer.



DATA SOURCES & PREPROCESSING

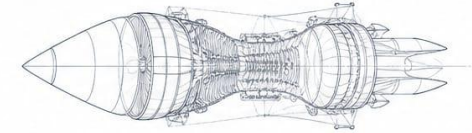
From Raw Sensor Logs to Physics-Aware Features

DATA SOURCES — OFFICIAL HACKATHON DATASET



PHYSICS-INFORMED FEATURE ENGINEERING

Derived from raw turbojet sensor measurements



	ENGINEERED FEATURE	PHYSICAL FORMULA	ENGINEERING PURPOSE
	Compressor Pressure Ratio	P_2 / P_{amb}	Measures compressor efficiency
	Compressor Temperature Rise	$T_2 - T_{amb}$	Estimates compression work
	Combustor Pressure Loss	$P_2 - P_3$	Detects combustion losses
	Turbine Pressure Ratio	P_4 / P_3	Evaluates turbine expansion
	Turbine Temperature Drop	$T_3 - T_4$	Indicates extracted turbine work
	Fuel-to-RPM Ratio	$Fuel\ Flow / RPM$	Represents fuel consumption efficiency
	Compressor Temperature Ratio	T_2 / T_{amb}	Normalized compressor heating
	Combustor Temperature Ratio	T_3 / T_2	Combustion performance indicator
	Pressure Recovery Ratio	P_3 / P_2	Measures combustor pressure recovery
	Overall Thermal Index	$(T_3 - T_4) / (Fuel\ Flow)$	Estimates overall thermal efficiency



Physics-guided features improve model **interpretability**, **robustness**, and **prediction accuracy** by embedding **thermodynamic relationships** into the learning process.



MODEL ARCHITECTURE / IMPLEMENTATION PLAN

From Prediction Engine to Interactive Dashboard

- Digital twin prediction engine**
Wraps the best-performing model for each target — Extra Trees, Ridge, CatBoost — together with SHAP explainability, uncertainty quantification, rule-based fault diagnosis, predictive maintenance, RUL estimation, and virtual sensors into a single inference engine.
- FastAPI backend**
Exposes the prediction engine as a REST API. A single primary endpoint, POST /predict, accepts sensor inputs and returns a structured JSON response covering health, performance, diagnostics, and maintenance guidance.
- React + Vite dashboard**
Consumes the API to render an interactive engineering interface — flight input panel, engine schematic, health gauges, performance cards, trend charts, and dedicated panels for fault diagnosis, maintenance, RUL, and virtual sensors.



END-TO-END REQUEST FLOW

Sensor Inputs → Physics-Informed Features → Best ML Models → Digital Twin Intelligence → JSON Response → React Dashboard

EXPECTED OUTCOMES & EVALUATION METRICS

How We'll Know AeroTwin Actually Works

1

System outputs

Six calibrated predictions — Compressor, Combustor, Turbine, and Overall Health, plus Thrust and TSFC — paired with prediction confidence and SHAP-based explanations for every output.

2

Decision-support layer

Fault diagnosis, maintenance recommendations, remaining useful life, and virtual sensor outputs turn raw predictions into something an engineer can act on, not just a number on a dashboard.

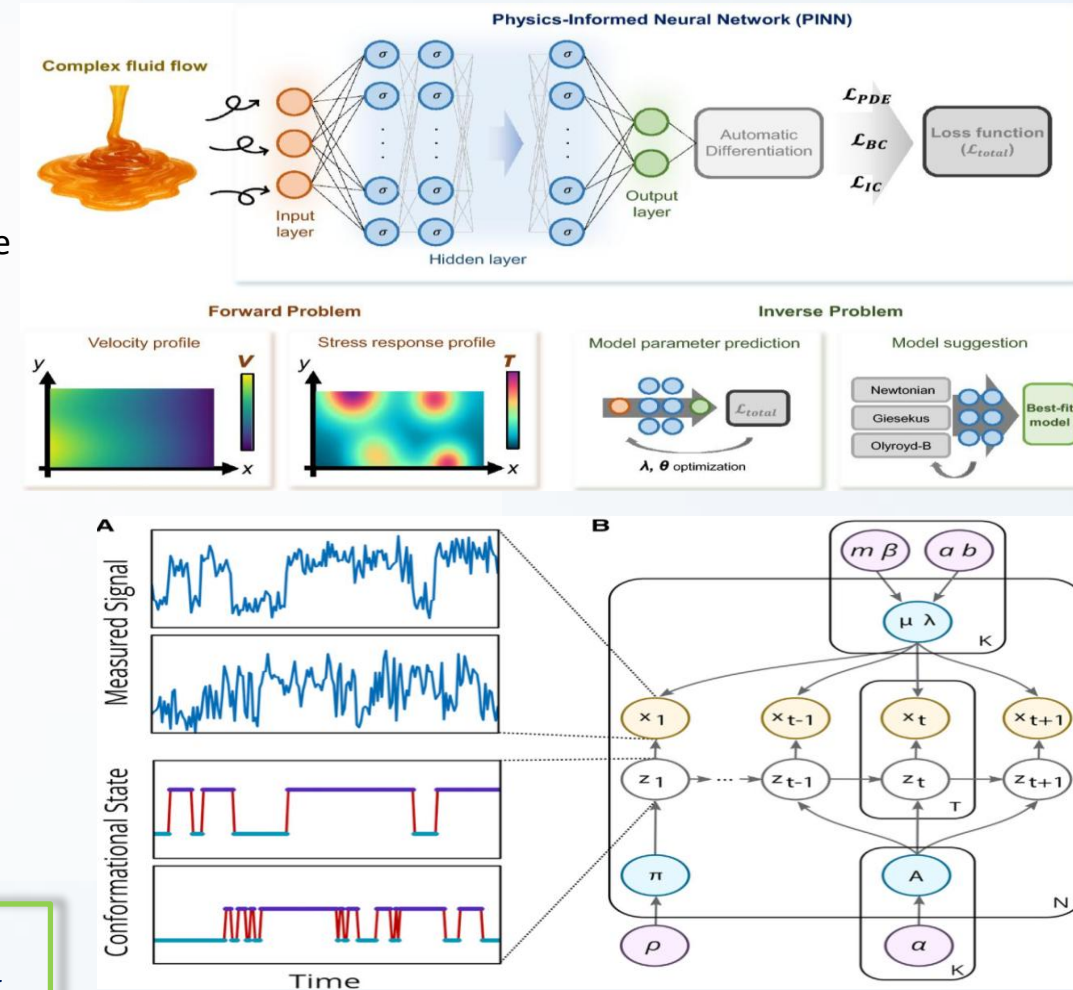
3

Model evaluation metrics

Every candidate model was scored on MAE, RMSE, R^2 , and training time during benchmarking — the same discipline used to select the best model per target is used to validate final performance.

SCOPE NOTE — REMAINING USEFUL LIFE

RUL is estimated using a health-threshold approximation, since the official dataset does not include explicit failure-cycle labels — flagged here rather than presented as ground-truth RUL.



TEAM COMPOSITION & CONTRIBUTIONS

💡 Team Name: VayuCops

- **Team Leader: Divyanshi**

Primary Contribution: ML Modelling and Benchmarking

GitHub: <https://github.com/Divyanshi12coder>

LinkedIn: www.linkedin.com/in/divyanshi-soni12

- **Member 1: Vishal Raj**

Primary Contribution: Fast API, Backend, Frontend

GitHub: <https://github.com/vishalrj291>

LinkedIn: <https://www.linkedin.com/in/vishal-raj-530a1b322>

- **Member 2: Vaishnavi Jaiswal**

Primary Contribution: Research and Full-Stack Pairing

GitHub: <https://github.com/vaishnaviii7557>

LinkedIn: <https://www.linkedin.com/in/vaishnavi-jaiswal-860248302>

MVP Link:

Frontend: <https://aerathon-2026-beta.vercel.app/>

Backend: <https://aerathon-2026.onrender.com>



APPENDIX & REFERENCES

💡 Sources, Tools, and the Complete Feature Set

• References

DATASET & PROBLEM STATEMENT

IIT Indore × HAL Hackathon 2026 — official problem statement and dataset: train.csv, test.csv, ground_truth.csv, turbojet_complete_dataset.csv

DOMAIN REFERENCE

Saravanamuttoo, Rogers, Cohen & Straznicky — Gas Turbine Theory (thermodynamic relations underlying physics-informed feature engineering)

ML & DATA LIBRARIES

Scikit-learn, XGBoost, LightGBM, CatBoost, SHAP, Pandas, NumPy, Matplotlib, Seaborn

BACKEND & FRONTEND

FastAPI, Uvicorn, Pydantic, Joblib · React, Vite, Axios, Recharts, Framer Motion, React Gauge Component

TOOLS & DEPLOYMENT

Git, GitHub, VS Code, Jupyter Notebook · Vercel

• APPENDIX A — COMPLETE PHYSICS-INFORMED FEATURE SET

Derived Feature	Computation
Compressor Pressure Ratio	P_2 / P_{amb}
Compressor Temperature Rise	$T_2 - T_{amb}$
Combustor Pressure Loss	$P_2 - P_3$
Turbine Pressure Ratio	P_4 / P_3
Turbine Temperature Drop	$T_3 - T_4$
Fuel Flow per Shaft Speed	Fuel Flow / RPM
Combustor Temperature Ratio	T_3 / T_2
Turbine Exit Temperature Ratio	T_4 / T_3
Compressor Pressure Recovery	P_3 / P_2
Turbine Expansion Ratio	P_4 / P_3

Click on the link below for Detailed Research:

https://drive.google.com/drive/folders/1Q_X2YKiXTX36IVvavkvnIjuKb1jxUn9w?usp=drive_link

